

54) As mentioned in the text, the notation  $\exists! x P(x)$  denotes "There exists a unique  $x$  such that  $P(x)$  is true." If the domain consists of all integers, what are the truth values of these statements?

a)  $\exists! x (x > 1)$ . This is False.  $2 > 1 \rightarrow \text{True}$   $3 > 1 \rightarrow \text{True}$ . So, it's not a unique  $x$ .

b)  $\exists! x (x^2 = 1)$ .  $x^2 = 1 \Rightarrow x^2 - 1 = 0 \Rightarrow (x-1)(x+1) = 0$ .  $\therefore x = 1$  or  $x = -1$ . So,  $x$  is not unique. This is False.

c)  $\exists! x (x + 3 = 2x)$   $x + 3 = 2x \Rightarrow x = 3$ . This is a unique  $x$ . So it's True.

d)  $\exists! x (x = x + 1)$   $x = x + 1 \Rightarrow 0 = 1$ . This is False.  $\therefore \exists x, (x = x + 1)$  is False since it's a contradiction for all  $x$ . There does not exist exactly one  $x$  s.t. the given expression is True.  $\therefore$  It's False.

55) What are the truth values of these statements?

a)  $\exists! x P(x) \rightarrow \exists x P(x)$   
 Let,  $\exists! x P(x)$  be True. By defn. of uniqueness quantifier, there exists exactly one element  $a$  in the domain such that  $P(a)$  is True.  $\therefore$  There exists at least one element  $a$  s.t.  $P(a)$  is True. By defn. of ~~uniqueness~~ <sup>existential</sup> quantifier,  $\exists x P(x)$  must be True. By defn. of conditional,  $\exists! x P(x) \rightarrow \exists x P(x)$  is True.

b)  $\forall x P(x) \rightarrow \exists! x P(x) \rightarrow$  If the hypothesis is False, then by defn. the <sup>cond is</sup> True  
 Let,  $\forall x P(x)$  be True. By defn. of universal quantifier, for all the elements  $a$  in the domain,  $P(a)$  is True. It's not the case that there is exactly one element for which  $P(a)$  is True. By defn. of uniqueness quantifier,  $\exists! x P(x)$  is False. By defn. of conditional,  $\forall x P(x) \rightarrow \exists! x P(x)$  is False.

c)  $\exists! x \neg P(x) \rightarrow \neg \forall x P(x)$   
 Let,  $\exists! x \neg P(x)$  be True. By defn. of uniqueness quantifier, there exists exactly one element  $a$  in the domain s.t.  $\neg P(a)$  is True.  $\therefore$  There exists at least one element  $a$  in the domain s.t.  $\neg P(x)$  is True. By defn. of existential quantifier,  $\exists x \neg P(x)$  is True. By De Morgan's law of quantifiers,  $\neg \forall x P(x)$  is True. By defn. of conditional,  $\boxed{\exists! x \neg P(x) \rightarrow \neg \forall x P(x)}$  is True.